Standard Deep Ensembles versus Concrete Dropout

Predictive Accuracy and Uncertainty for Gaussian Regression

Your Name CS 6956

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Abstract

This report compares standard deep ensembles with Concrete Dropout for heteroscedastic regression. Each standard ensemble member is independently initialized and trained with an explicit L_2 penalty, corresponding to a zero-mean Gaussian weight prior. Concrete Dropout instead learns dropout probabilities and uses stochastic forward passes from one fitted network. Experiments use a two-dimensional synthetic problem with known nonlinear mean and input-dependent noise, followed by the ten-feature scikit-learn diabetes data set. On the synthetic data, standard ensembles have slightly lower error and narrower intervals, while Concrete Dropout gives closer-to-nominal 90% coverage and substantially larger epistemic uncertainty. On diabetes, the two methods have similar test NLL and RMSE, but Concrete Dropout remains more conservative in its intervals. These results illustrate that matching the number of predictive components does not make independently trained networks and dropout masks equivalent sources of uncertainty.

1 Introduction

Deep neural networks can achieve good point-prediction accuracy while remaining overconfident away from the training distribution. Predictive uncertainty is useful both for identifying difficult examples and for distinguishing observation noise from uncertainty about the model itself. This work studies two practical approximate Bayesian strategies. A standard deep ensemble trains multiple networks from different random initializations and random seeds [1]. Concrete Dropout learns continuous dropout probabilities, allowing dropout to remain stochastic at test time [2].

The central question is: *when the number of predictive components is matched, how do the methods differ in accuracy, calibration, and the aleatoric/epistemic decomposition?* The experiment first uses a synthetic two-dimensional regression task, where the data-generating function and noise are known. It then evaluates the methods on a higher-dimensional, realistic regression problem. Both training and held-out test performance are reported.

2 Methods

2.1 Probabilistic regression model

Every network is a two-hidden-layer ReLU MLP with 64 units per layer. Its output is a mean and log variance,

$$p(y \mid x, W) = \mathcal{N}\big(y; \mu_W(x), \sigma_W^2(x)\big)\,,$$

trained with Gaussian negative log likelihood (NLL). The predicted observation variance is interpreted as aleatoric uncertainty. Variation in the predicted means across members or stochastic passes is interpreted as epistemic uncertainty.

For a standard ensemble of S members, the objective is

$$\mathcal{L}_{\mathrm{DE}} = \frac{1}{n}\sum_i \mathrm{NLL}(y_i \mid x_i, W) + \lambda \sum_l \|W_l\|_2^2, \qquad \lambda = 10^{-4}.$$

The second term is the negative-log-prior contribution of an isotropic Gaussian prior on the weights, up to a constant and scale convention. Members use distinct seeds and are trained independently.

Concrete Dropout replaces ordinary dropout with a differentiable relaxed Bernoulli mask and learns a dropout probability p_l for each hidden layer [2]. Its regularizer contains a scaled weight-prior term and the dropout entropy term

$$\sum_l d_l \left[p_l \log p_l + (1-p_l)\log(1-p_l)\right],$$

with entropy coefficient 10^{-3}. At evaluation time, the same network is run with stochastic masks. Thus, a comparison with S standard members uses S Concrete Monte Carlo passes, but these components are correlated draws from one learned model rather than independently trained models.

For component means μ_s and variances v_s, predictions are aggregated using the law of total variance:

$$\bar{\mu} = \frac{1}{S}\sum_s \mu_s, \quad v_{\mathrm{alea}} = \frac{1}{S}\sum_s v_s, \quad v_{\mathrm{epi}} = \mathrm{Var}_s(\mu_s), \quad v_{\mathrm{total}} = v_{\mathrm{alea}} +$$

Mixture NLL is evaluated with log-sum-exp over the component Gaussians. Nominal 90% intervals are estimated by sampling the Gaussian mixture, not by replacing it with a single Gaussian.

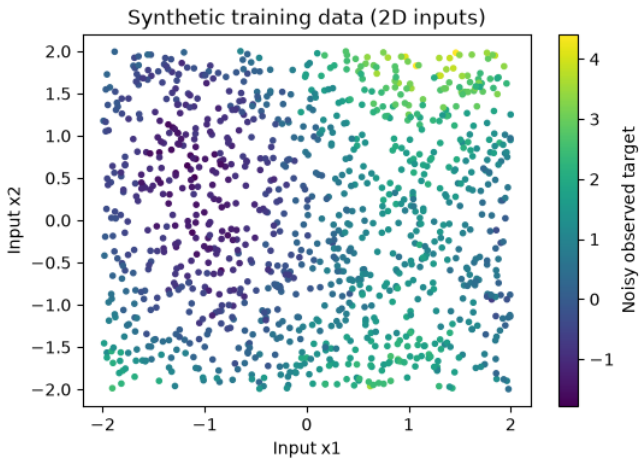


Figure 1: Synthetic training data. The two inputs are sufficient to visualize the full input space.

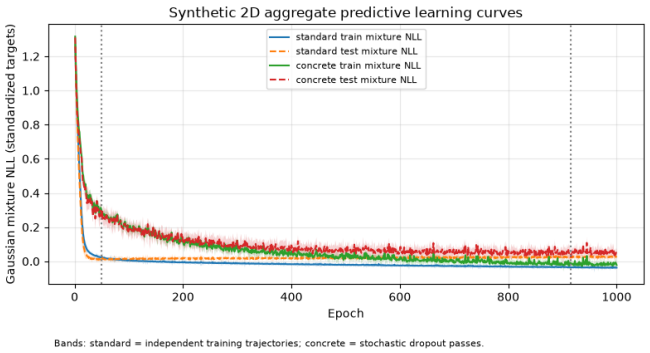


Figure 2: Synthetic aggregate predictive learning curves. Solid and dashed lines denote train and test mixture NLL, respectively.

2.2 Data and protocol

The synthetic data contain 1,500 points uniformly sampled from $[-2, 2]^2$. The noiseless mean is

$$f(x) = 1.3\sin(1.5x_1) + 0.45x_2^2 + 0.3x_1x_2,$$

and the standard deviation increases with both x_1 and $|x_2|$:

$$\sigma(x) = 0.18 + 0.13\frac{x_1 + 2}{4} + 0.12\frac{|x_2|}{2}.$$

The fixed split contains 1,125 training and 375 test points. An OOD grid from $[-3, 3]^2$ is used only for visualization. The realistic data are the 442-observation, ten-feature diabetes regression data set from scikit-learn, split into 331 training and 111 test points. Predictors and targets are standardized using training statistics; errors and interval widths are transformed back to original target units.

The full run trains ten standard members with seeds {11,23,37,51,71,83,97,109,127,139}. The Concrete model uses seed 11. Synthetic models are trained for 1,000 epochs and diabetes models for 20 epochs. Ensemble sizes from one through ten are evaluated. The main matched comparison uses M standard members versus M Concrete passes.

3 Results

3.1 Synthetic two-dimensional task

Figure 1 shows the noisy training observations. The known smooth mean and heteroscedastic noise make this a useful sanity check: uncertainty should increase where the observation noise is larger, and epistemic uncertainty should be more visible outside the square used for training.

The final per-model standardized test NLLs for the ten standard members range from 0.021 to 0.066, whereas Concrete Dropout obtains 0.085. The standard ensemble's best aggregate test NLL is 0.008 at epoch 50; the Concrete model's best is 0.026 at epoch 916 (Figure 2). This early standard-ensemble optimum is important: continuing to 1,000 epochs gives excellent fits but does not improve the aggregate held-out NLL.

Table 1: Synthetic test metrics in original target units.

Method	S	RMSE	NLL	Cov.	Width	SD_e
Standard	1	.305	.239	.891	.970	.000
Concrete	1	.310	.251	.899	1.016	.000
Standard	5	.305	.235	.877	.963	.034
Concrete	5	.306	.241	.915	1.037	.074

Table 1 reports representative one- and five-component comparisons. A single standard member has test RMSE 0.305 and 90% coverage 0.891. Five members leave RMSE nearly unchanged at 0.305, slightly reduce NLL to 0.235, and produce epistemic standard deviation 0.034. Concrete Dropout with five passes has test RMSE 0.306, NLL 0.241, coverage 0.915, and epistemic standard deviation 0.074. Hence, Concrete Dropout is closer to the nominal coverage target in this run, but its intervals are wider (1.037 versus 0.963 for standard ensembles). The extra width comes primarily from both larger aleatoric estimates and larger stochastic mean variation.

Increasing standard ensemble size changes little after three members: test RMSE remains approximately 0.304–0.305 and epistemic SD rises from 0.016 at two members to 0.038 at nine members. This is not a claim that the uncertainty estimate is monotonically converged; it reflects finite-sample estimation of member disagreement. Concrete MC passes improve the one-pass test NLL from 0.251 to 0.237 at two passes, then stabilize around 0.238–0.241. Epistemic SD rises from zero by construction to approximately 0.07–0.08.

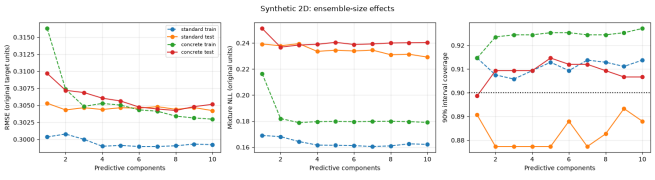


Figure 3: Synthetic RMSE, NLL, and 90% coverage as the matched number of predictive components increases.

3.2 Training versus test behavior and OOD uncertainty

Training metrics are better than test metrics for both methods. For example, the one-member standard model has train/test RMSE 0.300/0.305 and NLL 0.169/0.239. The five-member Concrete result has train/test RMSE 0.305/0.306 and NLL 0.180/0.241. The gap is modest on this smooth, well-sampled problem, but it is visible in likelihood and coverage.

The one-dimensional slice in Figure 4 follows $x_2 = 0$ and shows that both methods recover the mean function in the training domain. The OOD maps in Figure 5 visualize epistemic standard deviation outside $[-2, 2]^2$. Concrete Dropout produces stronger and more spatially diffuse disagreement. This is useful as an uncertainty signal, but it is not a guarantee of calibrated OOD probabilities: the OOD grid is synthetic and was not used to fit or tune either model.

3.3 Diabetes regression

The diabetes task is more data-limited and has ten correlated predictors. Standard members finish with test NLLs from 1.031 to 1.114 (standardized target units), while Concrete Dropout finishes at 1.052. The best aggregate test NLLs are 1.041 for standard ensembling and 1.022 for Concrete Dropout, at epochs 13 and 14, respectively. Thus, the method ranking changes slightly relative to the synthetic experiment when the model must infer a ten-dimensional relationship from only 331 training examples.

At five predictive components, standard ensembling gives test RMSE 54.423, NLL 5.392, coverage 0.883, interval width 159.355, and epistemic SD 7.045. Concrete Dropout gives RMSE 53.501, NLL 5.354, coverage 0.901, width 171.145, and epistemic SD 12.873. Concrete Dropout therefore has the best of these point and likelihood metrics in this particular split and is closer to 90% coverage, at the cost of wider intervals. The learned dropout rates are approximately 0.098 and 0.103, compared with 0.0037 and 0.0044 on the synthetic task. The difference suggests that the learned regularization responded to the much smaller, noisier effective data set, although more seeds and repeated splits would be needed to establish that conclusion.

4 Discussion and limitations

The experiments support three conclusions. First, standard deep ensembles are a strong accuracy baseline: independent optimization gives low synthetic error and narrow intervals without requiring learned dropout rates. Second, Concrete Dropout produces more epistemic variation and, in these runs, coverage nearer the nominal 90% level. This advantage is not free; the intervals are wider and the one-network stochastic samples are not equivalent to independent posterior samples. Third, the realistic-data ranking is less decisive. Concrete Dropout slightly improves the reported diabetes test metrics, but the data set is small enough that a single split and ten seeds do not support a broad claim of superiority.

The MC-pass experiment (Figure 8) separates the effect of sampling from the effect of fitting additional networks. On the synthetic test set, Concrete NLL improves from 0.251 at one pass to 0.237 at two passes and then largely plateaus. More passes mostly stabilize estimates of the mixture and epistemic variance rather than add new learned functions.

Important limitations are the Gaussian observation assumption, the hand-selected regularization strengths, the small number of independent seeds, and the use of one fixed train/test split for diabetes. Coverage is also an aggregate statistic; it does not establish conditional calibration across input regions. Finally, the L_2 term supplies a Gaussian-prior interpretation but neither method performs exact Bayesian inference.

5 Conclusion

For the synthetic problem, standard deep ensembles provide the best point-prediction and NLL results with relatively narrow intervals, while Concrete Dropout yields more conservative and better-covered intervals. On diabetes, Concrete Dropout is competitive or slightly better on the reported split, but with wider uncertainty. The practical choice therefore depends on whether the priority is computationally simple accuracy or a stronger uncertainty signal from a single learned stochastic model.

References

[1] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *NeurIPS*, 2017.

[2] Y. Gal, J. Hron, and A. Kendall. Concrete dropout. *NeurIPS*, 2017.

[3] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. *ICML*, 2016.

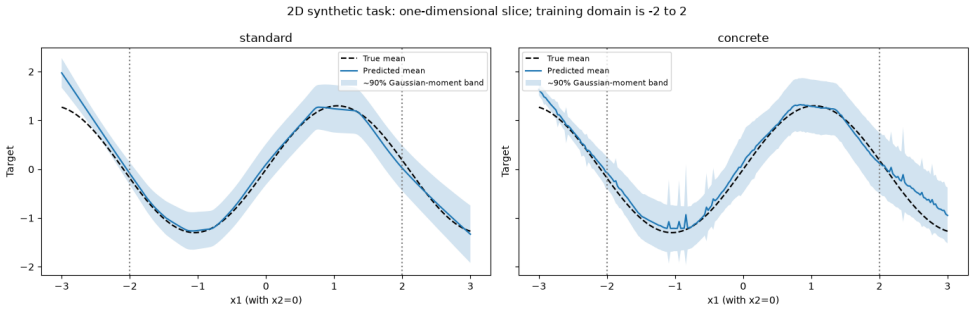


Figure 4: Predicted means and approximate 90% moment bands along $x_2 = 0$. Vertical dotted lines mark the training-domain boundary.

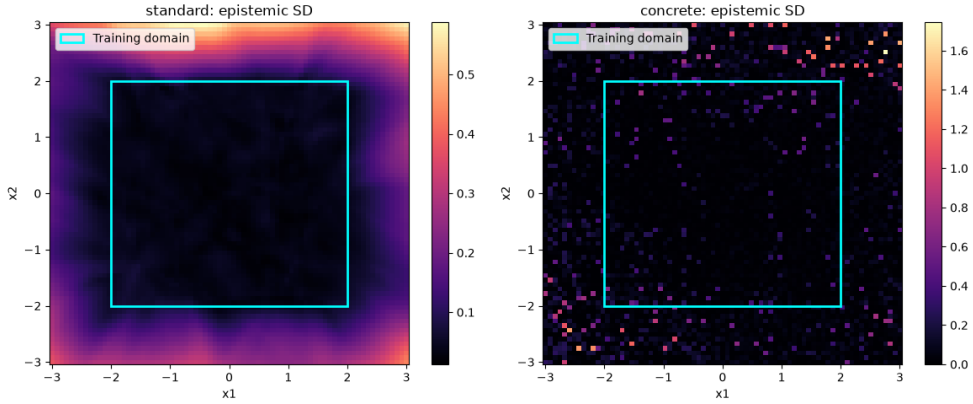


Figure 5: Epistemic standard deviation over the synthetic input plane. The cyan square is the training domain.

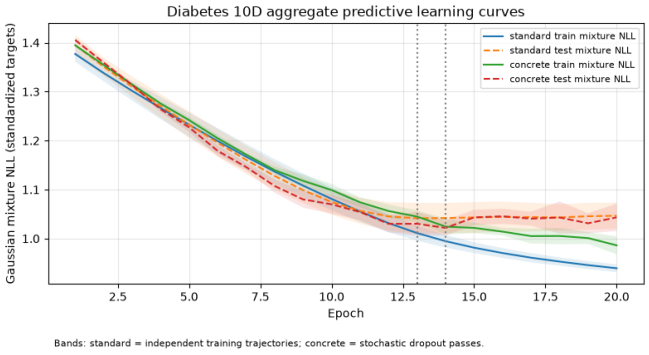


Figure 6: Diabetes aggregate predictive learning curves.

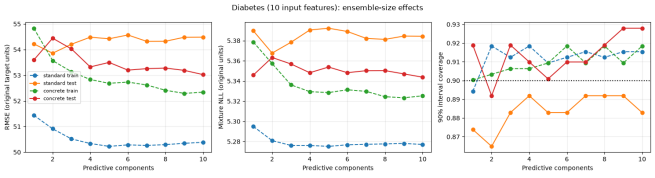


Figure 7: Diabetes RMSE, NLL, and 90% coverage versus predictive component count.

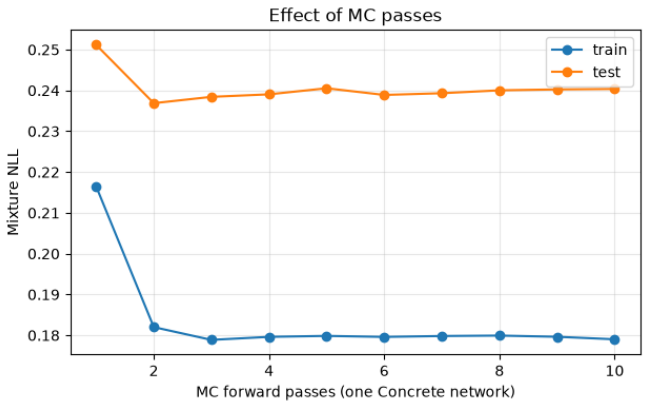


Figure 8: Sensitivity of Concrete Dropout test metrics to the number of stochastic forward passes.